

PART-I

$$(2) m_{eq} \frac{dv}{dt} = F_x - mg \sin \theta - f + g \cos \theta - 0.5 \rho A C_d (v + v_w)^2$$

Equivalent mass due to rotating parts;

$$E_{kin, trans} = \frac{1}{2} m v^2 \quad E_{kin, rot} = \frac{1}{2} \sum I_{cm} \omega_w^2 + \frac{1}{2} I_e \left(\frac{G, R}{r} v \right)^2$$

$$E_L = \frac{1}{2} \left[m + \frac{\sum I_{cm}}{r^2} + \frac{I_e (G, R)^2}{r^2} \right] v^2$$

$\underbrace{\hspace{10em}}_{m_{eq}}$

If we call additional term as 'd', $m_{eq} = m(1+d)$

(3) Linearizing NDE in (1) around point (θ_0, F_0, u_0) ;

Get Jacobians:

$$\frac{\partial f}{\partial u} = \frac{-\rho A C_d (u_0 + v_w)^2}{m_{eq} u}$$

$$\frac{\partial f}{\partial F} = \frac{1}{m_{eq} v} \quad \frac{\partial f}{\partial \theta} = \frac{m}{m_{eq} v} \cdot (-g \cos \theta_0 + f \sin \theta_0)$$

$$\dot{u} \cdot u' = \frac{\partial f}{\partial u} \cdot u' + \frac{\partial f}{\partial F} \cdot F' + \frac{\partial f}{\partial \theta} \cdot \theta'$$

$$\dot{u} \cdot u' = \frac{-\rho A C_d (u_0 + v_w)^2}{m_{eq} u} \cdot u' + \frac{1}{m_{eq} v} \cdot F' + \frac{m}{m_{eq} v} (-g \cos \theta_0 + f \sin \theta_0) \theta'$$

$$m_{eq} \cdot \frac{\dot{u} \cdot u'}{\rho A C_d (u_0 + v_w)} + u' = \frac{1}{\rho A C_d (u_0 + v_w)} \cdot F' + \frac{m}{m_{eq} v} \cdot \frac{(-g \cos \theta_0 + f \sin \theta_0)}{\rho A C_d (u_0 + v_w)} \cdot \theta'$$

Rearrange into standard form:

$$\boxed{\tau \ddot{u}' + u' = K(F' + d)}; \quad \boxed{\tau = \frac{m_{eq} v}{\rho A C_d (u_0 + v_w)}} \quad \boxed{K = \frac{1}{\rho A C_d (u_0 + v_w)}}$$

Input

(1)

$$\boxed{d = mg(-\cos \theta_0 + f \sin \theta_0)} \quad \text{Disturbance}$$

(3) (continued)

As depicted, F' is input and d is distance.

Following parameters are chosen for system:

$$u_0 = 40 \text{ m/s} \quad \theta_0 = 0 \quad m = 1250 \text{ kg} \quad \rho = 1.225 \text{ kg/m}^3$$

$$A = 1.2 \text{ m}^2 \quad Cd = 0.5 \quad f = 0.015 \quad u_w = 1.5 \text{ m/s}$$

$$\sum I_w = 4 \text{ kg.m}^2 \quad I_e = 0.15 \text{ kg.m}^2 \quad r = 0.3 \text{ m}$$

$$D = 3.0 \quad G = 1.1 \text{ (4th gear)}$$

$$m_{\text{eqv}} = m + \frac{\sum I_w}{r^2} + \frac{I_e (G \cdot r)^2}{r^2}$$

$$m_{\text{eqv}} = (1250) + \frac{4}{(0.3)^2} + \frac{(0.15) (1.1 \cdot 0.3)^2}{(0.3)^2}$$

$$m_{\text{eqv}} = 1313.1 \text{ kg}$$

$$\bar{t} = \frac{m_{\text{eqv}}}{\rho A C_d (u_0 + u_w)} = \frac{(1313)}{(1.225)(1.2)(0.5)(40 + 1.5)}$$

$$\bar{t} = 42.53 \text{ s}$$

PART-I

(5) For ICE, the input given to the system, (F_x), is throttle position.

For Electric motor, it is the electrical current.